

HARSHA KANDREGULA

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EXPERIENCE

Student Association Council Member

Aug. 2023 – Present

Vignan's Institute of Information Technology

Andhra Pradesh, India

- Orchestrated the planning and execution of **5+** university-wide technical and cultural events, driving engagement for **2,000+** students across **8** academic departments.
- Collaborated with a cross-functional team of **15+** committee members and university administration to manage logistics and marketing campaigns reaching **3,000+** students.
- Facilitated weekly communication between **10+** student bodies and **20+** faculty members, reducing event setup bottlenecks and saving **10+ hours** of administrative work per week.

PROJECTS

Nexus | FastAPI, PostgreSQL, Redis, AWS, Docker | GitHub

June 2026 – Present

- Developed an asynchronous FastAPI backend for a document Q&A platform, enabling users to upload PDFs (up to **50MB**) and process **10,000+** tokens per query using LLM integrations.
- Reduced API latency by **85%** (**800ms** → **120ms**) using Redis caching and content hashing to avoid reprocessing duplicate uploads.
- Improved system uptime to **99.9%** during third-party LLM outages through a circuit breaker pattern, preventing **100%** of cascading failures across **4** microservices.
- Maintained code reliability with strict type checking and **25+** integration tests that spun up PostgreSQL and S3 instances during GitHub Actions CI/CD runs.

Daemon | FastAPI, SQLAlchemy, LangChain, Plotly, Pandas, Docker | GitHub

May 2026 – Present

- Architected a voice-first self-regulation platform blocking **50+** distracting domains and serving **100+** beta testers; deployed via Docker Compose on Render.
- Integrated AssemblyAI achieving **95%** transcription accuracy for voice-input logging, reducing manual behavioral data entry by **10+ hours** per week.
- Engineered predictive analytics analyzing **5,000+** behavioral data points via scikit-learn, achieving an **R-squared of 0.85** in forecasting performance plateaus.
- Implemented strict data integrity with SQLAlchemy 2.0 ORM, explicit transaction boundaries, APScheduler background jobs, and **70%+** pytest coverage.

TECHNICAL SKILLS

Languages: Python, SQL, Bash

Backend: FastAPI, Pydantic, SQLAlchemy, Alembic, pytest, AsyncIO

Databases: PostgreSQL, Redis

AI / LLM: LangChain, OpenRouter API, tiktoken, OpenAI API, Anthropic API

Cloud & Infrastructure: AWS, Docker, GitHub Actions, LocalStack

Tools: Git, Linux, Postman, uv, ruff, mypy

EDUCATION

Vignan's Institute of Information Technology

Andhra Pradesh, India

B.Tech in Electrical and Electronics Engineering | **CGPA: 8.6/10**

Aug. 2023 – May 2027

ACHIEVEMENTS

Winner – SUSHACKS 2025 | Project: Verdex

Mar. 2025

- Outperformed **50+** competing teams to deliver an AI energy transparency solution during an intensive **24-hour hackathon**, visualizing real-time carbon footprints across **10+** LLM providers.

Winner – MINDCRAFTERS 2024 | Project: ClassroomCritique

Nov. 2024

- Shipped a production-ready anonymous faculty rating system within a **24-hour coding sprint**, architecting the database to scale for **5,000+** student users and **200+** faculty profiles.
- Designed a department-level scoring algorithm processing **15+** academic metrics to optimize college selection, improving recommendation accuracy by **30%**.